On the Structure of Portable Documents

The portable document format describes pages as a sequence of drawing operators applied to a graphics state. Text is not stored as paragraphs but as positioned glyph runs, which is precisely what makes **in-place editing** a hard problem. Every editor that claims otherwise is drawing *white rectangles*.

Content streams

A content stream interleaves path construction, color changes and text objects. The operator `Tj` shows a string, while `TJ` accepts an array mixing strings with kerning adjustments. Fonts are referenced by name through the resource dictionary and are usually embedded as subsets.

Cross reference data

Modern writers emit cross reference streams with three column widths and pack non-stream objects into object streams. A robust parser therefore needs the full filter pipeline before it can even locate the document catalog. This paragraph exists mainly to produce several distinct text runs across multiple lines, with enough length to trigger line wrapping in the layout engine.